

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Regular/Supplementary Winter Examination – 2024****Course: B. Tech. Branch : Electronics/ Electronics and Telecommunication/****Electronics and Communication Engineering****Semester : III****Subject Code & Name: BTETC302/ BTEXC302 & Electronic Devices and Circuits****Max Marks: 60****Date: 07/02/2025****Duration: 3 Hr.****Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
	Which region of the transistor is highly doped?	CO1	1
	a. Emitter b. Base c. Collector d. Both Emitter & Collector		
	What is the use of coupling capacitors in CE amplifier?	CO2	1
	a. Block DC b. Pass AC c. Reduce distortion d. Pass AC and Block DC		
	What is the correct phase shift in CE amplifier?	CO1	1
	a. 45 Degrees b. 60 Degrees c. 120 Degrees d. 180 Degrees		
	What is the main advantage of FET which makes it more useful in industrial applications?	CO2	1
	a. Voltage controlled operation b. Less cost c. Small size d. Semiconductor device		
	N channel Depletion MOSFET is known as	CO1	1
	a. Normally ON then OFF b. Normally OFF then ON c. Normally ON d. Normally OFF		
	P channel Enhancement MOSFET is known as	CO2	1
	a. Normally ON then OFF b. Normally OFF then ON c. Normally ON d. Normally OFF		
	Power amplifier directly amplifies _____	CO1	1

	a. Voltage of signal	b. Current of the signal	c. Power of the signal	d. All of the mentioned		
8	Where does the Q point lie for class B amplifier?				CO2	1
	a. Active	b. Cut off	c. Saturation	d. Between saturation and active		
9	Bandwidth due to negative feedback amplifier				CO2	1
	a. Increases	b. Decreases	c. Remains same	d. Partially changes		
10	Gain due to negative feedback amplifier				CO1	1
	a. Increases	b. Decreases	c. Remains same	d. Partially changes		
11	According to Barkhausen criteria of oscillation total phase shift should be-----Degrees				CO1	1
	a. 0 or 360	b. 180	c. 270	d. 90		
12	Which pin number is the input for 7905 IC				CO2	1
	a. 1	b. 2	c. 3	d. 4		
Q. 2	Solve the following.					12
A)	Explain need of biasing and voltage divider biasing method				CO1	6
B)	Explain in detail effects of bypass and coupling capacitor				CO2	6
Q.3	Solve the following.					12
A)	Explain JFET and its characteristics, explain the FET as VVR.				CO2	6
B)	Explain in detail D MOSFET and E MOSFET				CO1	6
Q. 4	Solve Any Two of the following.					12
A)	Explain class A power amplifier and derive expression for efficiency				CO2	6
B)	Explain class B push pull power amplifier and derive expression for efficiency				CO1	6
C)	Explain crossover distortion and need of heat sink, its design				CO1	6
Q.5	Solve Any Two of the following.					12
A)	Derive expression for voltage gain with feedback for both positive as well as negative feedback amplifier				CO3	6
B)	Explain with block diagram types of feedback amplifier				CO1	6

C)	Explain effect of negative feedback amplifier on input, output impedance, voltage gain, current gain, bandwidth, noise and distortion	CO2	6
Q. 6	Solve Any Two of the following.		12
A)	Explain Barkhausen criteria of oscillation and RC phase shift oscillator	CO1	6
B)	Explain and derive expression for oscillation frequency of Wien bridge oscillator	CO2	6
C)	Explain in detail transistorized series voltage regulator	CO2	6
*** End ***			

9DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

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Semester : III

Branch : Electronics Engineering/ Electronics and Telecommunication Engineering

Subject Code & Name: Electronic Devices & Circuits/ BTEXC303_Y18

Max Marks: 60

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Q. 1 Objective type questions. (Compulsory Question)

- | | (Level/CO) | Marks |
|---|------------|-------|
| 1 What does JFET stand for? | | 12 |
| a) Junction Field-Effect Transistor. | CO1 | 1 |
| b) Junction Frequency-Effect Transistor. | | |
| c) Junction Forward-Effect Transistor. | | |
| d) Junction Feedback-Effect Transistor. | | |
| 2 What is the pinch-off voltage in a JFET? | CO1 | 1 |
| a) The voltage at which the channel is fully closed and no current flows. | | |
| b) The voltage at which the JFET starts conducting. | | |
| c) The voltage at which the JFET breaks down. | | |
| d) The voltage at which the JFET enters the saturation region. | | |
| 3 What is the purpose of biasing in a JFET? | CO2 | 1 |
| a) To set the operating point in the desired region. | | |
| b) To increase the gain of the JFET. | | |
| c) To reduce the input impedance. | | |
| d) To decrease the output impedance. | | |
| 4 What is the primary function of the gate in a MOSFET? | CO3 | 1 |
| a) To control the flow of current between the source and drain. | | |
| b) To provide power to the MOSFET. | | |
| c) To act as a heat sink. | | |
| d) To connect the MOSFET to the ground. | | |

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|----|---|-----|---|
| 5 | What is the threshold voltage in an E-MOSFET? | CO3 | 1 |
| | a) The minimum gate-to-source voltage required to form a conductive channel. | | |
| | b) The maximum drain-to-source voltage. | | |
| | c) The voltage at which the MOSFET breaks down. | | |
| | d) The voltage at which the MOSFET enters the saturation region. | | |
| 6 | In which region does a MOSFET operate when used as a switch? | CO3 | 1 |
| | a) Cut-off region. | | |
| | b) Ohmic region. | | |
| | c) Saturation region. | | |
| | d) Breakdown region. | | |
| 7 | Which of the following amplifiers is designed to amplify weak signals with minimal addition of noise? | CO4 | 1 |
| | a) Power Amplifiers | | |
| | b) Low Noise Amplifiers | | |
| | c) Voltage Amplifiers | | |
| | d) Operational Amplifiers | | |
| 8 | What is the primary purpose of a power amplifier? | CO4 | 1 |
| | a) To amplify weak signals | | |
| | b) To drive loudspeakers or high-power loads | | |
| | c) To filter noise | | |
| | d) To provide high-frequency gain | | |
| 9 | Which of the following oscillators is most commonly used for audio frequency generation? | CO4 | 1 |
| | a) Hartley Oscillator | | |
| | b) Colpitts Oscillator | | |
| | c) Wein Bridge Oscillator | | |
| | d) Crystal Oscillator | | |
| 10 | The Barkhausen criterion is used in the analysis of which type of circuits? | CO4 | 1 |
| | a) Amplifiers | | |
| | b) Oscillators | | |
| | c) Filters | | |
| | d) Attenuators | | |
| 11 | What is the main feature of a monostable multivibrator? | CO5 | 1 |
| | a) It has two stable states and produces a square wave. | | |
| | b) It has one stable state and produces a single pulse when triggered. | | |
| | c) It has no stable states. | | |
| | d) It operates with a fixed frequency. | | |

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|----|---|-----|---|
| 12 | A Switch Mode Power Supply (SMPS) is more efficient than a linear power supply because: | CO5 | 1 |
| | a) It generates no heat. | | |
| | b) It uses a low-frequency transformer. | | |
| | c) It switches on and off, reducing power loss. | | |
| | d) It operates without feedback. | | |

Q. 2 Solve the following.

- | | | | |
|----|---|-----|----|
| A) | Determine I_{DQ} , V_{GSQ} , V_D , V_S , V_{DS} and V_{DG} for the network of figure 1. | CO1 | 12 |
| | | | 6 |

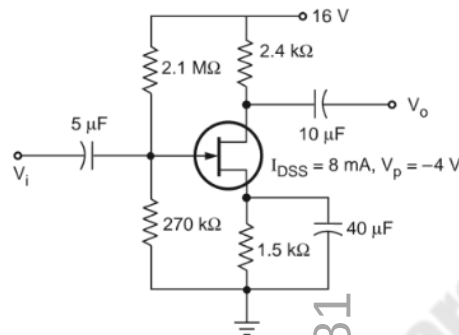


Figure 1

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|----|--|-----|---|
| B) | Compare BJT, FET and MOSFET with different parameters. | CO1 | 6 |
|----|--|-----|---|

Q.3 Solve the following.

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|----|---|---------|----|
| A) | Explain the drain characteristics and transfer characteristics of N-channel depletion type MOSFET. | CO1 | 12 |
| | | | 6 |
| B) | Determine the d.c bias point for the EMOSFET circuit shown in figure 2. Assume $k=0.4\text{mA/V}^2$, $V_T=3\text{V}$. | CO1/CO2 | 6 |

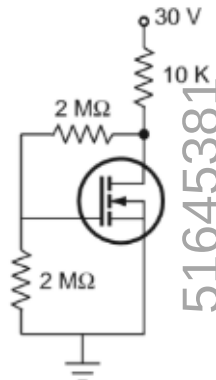


Figure 2

Q. 4 Solve Any Two of the following.

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|----|--|-----|----|
| A) | With a neat circuit diagram explain the working of a MOSFET as a switch. | CO3 | 12 |
| | | | 6 |

- | | | | |
|----|---|-----|---|
| B) | Explain the classification of power amplifiers in detail. | CO3 | 6 |
| C) | What is a feedback amplifier? Explain the types of feedback amplifiers. | CO3 | 6 |

Q.5 Solve Any Two of the following. 12

- | | | | |
|----|---|---------|---|
| A) | With a neat circuit diagram derive the expression of frequency of oscillations for a Colpitts oscillator. | CO3/CO4 | 6 |
| B) | State and explain Barkhausen criterion of Oscillations. | CO4 | 6 |
| C) | Explain the concept of frequency stability of Oscillators. | CO4 | 6 |

Q. 6 Solve Any Two of the following. 12

- | | | | |
|----|---|-----|---|
| A) | What is a multivibrator. Explain the operation of monostable multivibrator using IC555 timer. | CO4 | 6 |
| B) | With a neat block diagram explain the IC555 timer. | CO5 | 6 |
| C) | Draw the block diagram of SMPS and also describe its operation. | CO5 | 6 |

*** End ***